5	**(a)**	curved line showing decreasing gradient with temperature rise	M1	
		smooth line not touching temperature axis, not horizontal or vertical anywhere	A1	[2]
	(b) **(i)**	(no energy lost in battery because) no/negligible internal resistance	B1	[1]

(ii) $I = V/R$

$= 8/15 \times 10^3$ or $1.6/3.0 \times 10^3$ or $2.4/4.5 \times 10^3$ or $12/22.5 \times 10^3$ C1

$= 0.53 \times 10^{-3}$ A A1 [2]

(iii) p.d. across X = $12 - 8.0 - 3.0 \times 10^3 \times 0.53 \times 10^{-3}$ (= 2.4 V) C1

$R_X = 2.4/(0.53 \times 10^{-3})$ C1

or
$R_{tot} = 12/0.53 \times 10^{-3}$ (= $22.5 \times 10^3\ \Omega$) (C1)
$R_X = (22.5 - 15.0 - 3.0) \times 10^3$ (C1)

$4.5(2) \times 10^3\ \Omega$ A1 [3]

(iv) resistance decreases hence current (in circuit) is greater M1

p.d. across X and Y is greater hence p.d across Z decreases A1

or explanation in terms of potential divider:
R_Z decreases so $R_Z/(R_X + R_Y + R_Z)$ is less (M1)
therefore p.d. across Z decreases (A1) [2]